

Microbiological Analysis Report

Facility: Demo Foods Inc. | Product: Baby spinach | Date: 06/09/2026 | Analyst: Pedro Antonio Pérez Ferrer

Overall risk level: **CRITICAL**

Executive Summary

Testing of three baby spinach lots at Demo Foods Inc. revealed one critical food safety failure: Salmonella was detected in LOT-003, making it an adulterated product that must be immediately quarantined and cannot be released under any circumstances. LOT-002 also showed elevated total coliform counts suggesting a sanitation or cross-contamination issue that requires prompt investigation. LOT-001 is the only fully compliant lot and may proceed through normal release procedures pending corrective actions on the broader facility concerns.

Analytical Results

Sample	Microorganism	Result	Limit	Status
LOT-001	Total coliforms	45 CFU/g	100 CFU/g	OK
LOT-001	Aerobic mesophiles	82,000 CFU/g	100000 CFU/g	OK
LOT-002	Total coliforms	320 CFU/g	100 CFU/g	WARNING
LOT-002	E. coli O157:H7	0 CFU/g	0 (absence) CFU/g	OK
LOT-003	Salmonella spp.	1 CFU/25g	0 (absence) CFU/25g	CRITICAL
LOT-003	Molds and yeasts	800 CFU/g	1000 CFU/g	OK

Required Corrective Actions

Sample LOT-001: Both total coliforms (45 CFU/g) and aerobic mesophile counts (82,000 CFU/g) are within acceptable limits. This lot appears microbiologically sound and poses no immediate food safety concern. The aerobic plate count, while compliant, is moderately elevated and worth monitoring as a trend indicator for product shelf life and incoming material quality.

- Approve LOT-001 for release following standard quality review procedures.
- Document and trend the aerobic mesophile count (82,000 CFU/g) to establish baseline and detect any future degradation in raw material quality or cold chain performance.
- Maintain current wash and sanitation parameters for this production run as a reference benchmark.

Sample LOT-002: Total coliform count of 320 CFU/g exceeds the acceptable limit and is flagged as a WARNING. Although the more dangerous E. coli O157:H7 was not detected (0 CFU/g), elevated total coliforms in ready-to-eat baby spinach is a significant indicator of fecal contamination, inadequate washing, sanitation breakdown, or post-process cross-contamination. This lot should be placed on hold pending investigation.

- Place LOT-002 on immediate quarantine hold — do not ship or release until root cause investigation is complete.
- Audit wash water sanitizer concentration logs for the LOT-002 production window; verify free chlorine or peracetic acid levels met process specifications throughout the run.
- Swab and test all food-contact surfaces, conveyor belts, flumes, and cutting equipment used during LOT-002 production for indicator organisms.
- Review worker hygiene records and conduct a GMP walkthrough for the production shift associated with LOT-002.
- Perform a water quality test on the wash water source and any recirculated water used during this production period.

- Consider microbiological retesting of LOT-002 with a larger sample plan before making a final disposition decision.
- Notify the QA manager and initiate a formal Non-Conformance Report (NCR) per the facility's food safety plan.

Sample LOT-003: Salmonella spp. was detected at 1 CFU/25g in LOT-003. Under FDA regulations and FSMA, any detectable Salmonella in a ready-to-eat product such as baby spinach constitutes adulteration — there is zero tolerance. This result is a critical food safety failure requiring immediate action. The mold and yeast count (800 CFU/g) is within acceptable range and is not a concern in this context, but it does not mitigate the Salmonella finding in any way.

- IMMEDIATELY quarantine and place a full hold on LOT-003 — this product must not be shipped, sold, or used under any circumstances.
- Notify senior management, the food safety team, and legal/regulatory affairs immediately.
- Initiate a full environmental monitoring sweep of the entire production facility, including drains, floors, walls, equipment surfaces, and hard-to-clean areas, testing specifically for Salmonella.
- Halt production on the affected line until environmental investigation is complete and the facility is verified Salmonella-free.
- Trace back LOT-003 raw materials to the grower or field source and conduct a supply chain investigation; notify the raw material supplier.
- Review and test all irrigation water sources associated with the implicated growing fields.
- Assess whether any product from the same production window or from overlapping raw material lots has already shipped; if so, initiate a mock recall or actual recall assessment in accordance with the facility's recall plan.
- Notify the FDA under FSMA Section 414 (recordkeeping and access) obligations and evaluate mandatory reporting requirements under 21 CFR Part 16 and the Reportable Food Registry (RFR) if the product has entered commerce.
- Destroy or divert LOT-003 to a non-human food use under regulatory oversight; document disposition.
- Conduct a full root cause analysis (RCA) and update the facility's Hazard Analysis and Food Safety Plan to address identified gaps before resuming production.

Regulatory Observations (FSMA)

- FSMA — Produce Safety Rule (21 CFR Part 112): Detection of Salmonella in LOT-003 is a direct violation of the zero-tolerance standard for human pathogens on ready-to-eat produce. Demo Foods Inc. must evaluate obligations under the Produce Safety Rule regarding agricultural water quality, biological soil amendments, and worker hygiene, and must document corrective actions in their food safety records.
- FSMA — Preventive Controls for Human Food (21 CFR Part 117): The Salmonella finding in LOT-003 indicates a potential failure of one or more preventive controls identified in the facility's Food Safety Plan. A reanalysis of the hazard analysis for Salmonella on baby spinach is required, and corrective action records must be generated and retained per 21 CFR 117.150.
- FDA Reportable Food Registry (RFR) — 21 CFR Part 16 / Section 417 of FD&C; Act: If LOT-003 has entered commerce (i.e., left the facility's control), Demo Foods Inc. has a legal obligation to submit a report to the FDA Reportable Food Registry within 24 hours of confirming a reasonable probability that the product will cause serious adverse health consequences. An internal assessment of distribution must be conducted immediately.
- FSMA — Supply Chain Program: The detection of Salmonella in incoming raw agricultural commodities suggests the facility's supplier verification program may be inadequate. Under 21 CFR Part 117 Subpart G, Demo Foods Inc. must apply appropriate supplier verification activities (e.g., onsite audits, microbiological testing) for raw materials that present a known pathogen hazard.
- FSMA — Recall Plan Requirement (21 CFR 117.139): A validated recall plan must be on file and must be activated or evaluated for activation given the LOT-003 Salmonella finding. Records of lot traceability, distribution records, and customer notifications must be prepared in accordance with the facility's written recall procedures.
- FSMA — Records and Recordkeeping (21 CFR 117.190): All corrective actions, hold decisions, environmental test results, and disposition records for LOT-002 and LOT-003 must be documented and retained for a minimum of 2 years and made available to FDA inspectors upon request.

Critical Control Point Recommendations

- **CCP-1 — Wash Water Sanitizer Control:** Establish or reinforce a validated Critical Control Point for wash water sanitizer concentration (e.g., free chlorine 50–200 ppm or peracetic acid per validated range) with defined critical limits, continuous or per-lot monitoring, corrective action procedures, and verified recordkeeping. The LOT-002 coliform exceedance strongly suggests this CCP may have failed or is not adequately defined.
- **CCP-2 — Incoming Raw Material Microbial Acceptance:** Establish a CCP or prerequisite program for incoming raw agricultural commodity (RAC) acceptance that includes Salmonella testing or supplier verification for high-risk growing regions, particularly for ready-to-eat leafy greens. Certificate of Analysis (COA) review alone is insufficient given the LOT-003 finding.
- **CCP-3 — Environmental Monitoring Program (EMP) for Salmonella:** Implement or enhance a zone-based Environmental Monitoring Program targeting Salmonella indicator organisms and Salmonella spp. in the production environment. Establish trigger thresholds, corrective action protocols, and trend analysis to detect harborage sites before product contamination occurs.
- **CCP-4 — Water Quality Control Point:** Designate water used in washing, flume transport, and equipment sanitation as a critical prerequisite control with verified testing for microbial indicators (coliforms, E. coli) on a defined frequency. Any water quality exceedance should trigger immediate corrective action and lot hold.
- **CCP-5 — Post-Process Cross-Contamination Prevention:** Establish strict physical and procedural separation between raw incoming product zones and post-wash ready-to-eat zones. Implement validated sanitation procedures with pre-operational and operational verification swabbing to prevent recontamination after the wash step.

FDA Recall Intelligence

Source: FDA Recall Enterprise System (RES) — open.fda.gov/apis/food/enforcement

Total coliforms FDA Ref: H-0579-2025	Total recalls on record: 37 Last 24 months: 2 Most recent (August 2025): Hershey s Ice Cream, Green Mint Chip (GMC) Bulk, 3 gallons, 1 unit , bulk can... Recalling firm: HERSHEY CREAMERY CO. Classification: Class II — avg recall cost: \$2.1M Verify at: accessdata.fda.gov/scripts/ires → search recall number above
Molds and yeasts FDA Ref: H-0025-2026	Total recalls on record: 177 Last 24 months: 11 Most recent (September 2025): Organic BABY bedtime drops; sleep + immunity blend; Promotes restful sleep wi... Recalling firm: M.O.M Enterprises, LLC. Classification: Class II — avg recall cost: \$2.1M Verify at: accessdata.fda.gov/scripts/ires → search recall number above
E. coli O157:H7 FDA Ref: H-0764-2026	Total recalls on record: 124 Last 24 months: 8 Most recent (April 2026): RAW FARM RAW CHEDDAR SIMPLY SHREDDED Original MADE WITH: WHOLE RAW MILK TRULY... Recalling firm: Raw Farm LLC Classification: Class I — avg recall cost: \$8.5M Verify at: accessdata.fda.gov/scripts/ires → search recall number above
Salmonella spp. FDA Ref: H-0835-2026	Total recalls on record: 100 Last 24 months: 15 Most recent (May 2026): Good & Gather Mexican Street Corn Trail Mix 8 oz bag UPC 085239270240, 8 bags... Recalling firm: JOHN B SANFILIPPO & SONS INC Classification: Class I — avg recall cost: \$8.5M Verify at: accessdata.fda.gov/scripts/ires → search recall number above

Report generated on 06/09/2026 at 19:25 by Pedro Antonio Pérez Ferrer. This analysis was AI-assisted and must be reviewed by a certified microbiologist before any regulatory decisions are made. Input data is processed locally and not stored or retained after report generation.